

M14 — Cohort debrief: data aggregation (PEARLS scaffold)

Phase: 9 — Debrief **Status:** DONE (2026-05-26) **Blocked by:** M10, M12, M13 **Blocks:** M15 **Estimated effort:** 2 days · **Actual:** 0.4 day

1. Purpose

Roll up every encounter in a ControlRoom into one PEARLS-scaffolded debrief JSON so the instructor can run a single debrief covering the whole cohort. PEARLS (Eppich & Cheng 2015) is the standard nursing-debrief framework: Reactions → Description → Analysis → Application → Summary. M14 builds the data; M15 renders it.

Robust to:

- An encounter with no transcript / no chart events.
- A completely empty room (operator started + immediately ended).
- Private-clone rooms (M13) — each clone is its own facet.
- An encounter whose per-encounter `build()` throws — the error surfaces as a `_error` field on that facet without breaking the aggregate.

2. Structure

Files touched:

- `portal/debrief.py` — appends "V7 — Cohort debrief (M14)" section:
- `build_cohort_debrief(room)` -> dict
- `save_cohort(debrief)` -> Path
- `load_cohort(room_id)` -> dict | None
- `list_saved_cohorts()` -> list[dict]
- Module constant `COHORT_DEBRIEFS_DIR = data/debriefs/cohort/`.

No new files — extends `portal/debrief.py`.

3. Uses

- **M15** will add `/portal/debrief/cohort/<room_id>` to render this JSON as a web page with PEARLS-section tabs.
- `POST /api/room/end` (a future enhancement) should call `build_cohort_debrief(room)` and `save_cohort(...)` BEFORE `end_active_room()` clears the singleton, so the room's in-memory state is captured. M15 will wire this.
- **Operator-driven retro debrief:** `load_cohort(room_id)` lets the instructor revisit any saved cohort debrief.

4. Functions (exported API surface)

Cohort builder

`build_cohort_debrief(room)` -> dict — `room` is a `control_room.ControlRoom`. Returns:

```
{
  "room_id":      "...",
  "room_code":    "ABCDEF",
  "room_label":   "...",
  "room_status":  "active|frozen|ended",
  "encounters":   [<full v6 per-encounter debrief>, ...],
  "pearls": {
    "reactions":  {"prompt": "...", "notes": ""},
    "description": {"prompt": "...", "facts": [<strings>]},
    "analysis":   {
      "prompt": "...",
      "performance_frames": [
        {session_id, scenario_name, ncjmm_coverage,
         duration_seconds, turns},
        ...
      ],
      "persona_engagement_ranked": [{persona_id, turn_count}, ...]
    },
    "application": {"prompt": "...", "commitments": []},
    "summary": {
      "encounters_count", "students_count",
      "total_chat_turns", "total_chart_events",
      "total_duration_seconds",
      "avg_duration_per_encounter_seconds",
      "avg_turns_per_encounter"
    }
  },
  "_meta": {"generated_at": <ts>, "generator": "medsim7.debrief.cohort v1"}
}
```

Persistence

- `save_cohort(debrief)` writes `data/debriefs/cohort/<room_id>.json`.
- `load_cohort(room_id)` returns the dict or `None`.
- `list_saved_cohorts()` returns the index (newest first) for an eventual cohort-debrief library page.

5. Limitations

- **No per-student facet rollup.** In a shared-mode encounter with 3 students, the facet is one debrief covering all 3 (matches v6 per-encounter behavior). To get per-student grouping, use `private_clone` mode (M13) — each clone produces its own facet.
- ****`reactions.notes` and `application.commitments` are empty at build time**** — the instructor fills them live during the debrief. M15 should provide UI for these.
- ****Persona engagement ranking is empty when there's no chat activity.**** M15 should hide the section in that case.
- **No PDF export.** A v7.1 enhancement.

- **Per-encounter** `build()` errors are silently captured in the

facet's `_error` field — they don't fail the aggregate but they do degrade the facet's content. M15 should surface the error prominently when rendering.

6. Test status

Test file	Cases	Status	Last run
<code>test_cohort_debrief_aggregates_3_encounters.py</code>	2 — 3-encounter aggregation with mixed chart activity; save/load round-trip to <code>data/debriefes/cohort/</code> .	PASS	2026-05-26
<code>test_cohort_debrief_includes_pearls_sections.py</code>	2 — All 5 PEARLS sections present with prompts/facts/frames; persona engagement section structure intact even when empty.	PASS	2026-05-26
<code>test_cohort_debrief_handles_one_encounter_with_no_chartin_g.py</code>	3 — Empty encounter handled with zeroed facet; completely empty room produces valid debrief; private-clone room produces one facet per clone.	PASS	2026-05-26

7/7 PASS. **Full v7 suite: 98/98 passing** (up from 91 — +7 M14 tests). **Full v6 regression on v7: 209 passed**, same 6 env-flaky pre-existing failures, **0 v7 regressions**.

7. Change list

Date	Author	Change	Files
2026-05-26	claude-code	Appended cohort-debrief section to <code>portal/debrief.py</code> : <code>build_cohort_debrief</code> + <code>save_cohort</code> + <code>load_cohort</code> + <code>list_saved_cohorts</code> + <code>COHORT_DEBRIEFS_DIR</code> . PEARLS scaffold pre-populates Description + Analysis from cohort facts; Reactions + Application empty for live fill. 7 acceptance tests across 3 files.	<code>portal/debrief.py</code> , <code>tests/v7/test_cohort_debrief_*.py</code>

8. Open questions / known issues

- **No instructor-side "complete this debrief" affordance.** Once

M15 lands, an instructor will be able to add notes to Reactions and commitments to Application; those edits should be saved back to the cohort JSON. M15 will own this.

- `encounters[]` **facet is the full v6 debrief — large payload.**

A 6-encounter room of 8 students could produce >100 KB of JSON. Acceptable for current scale; if the cohort grows, paginate or add a "summary mode" toggle.

- **Private_clone aggregation:** all clones share `cloned_from_id`.

M15 should visually group clones under their template card so the instructor sees "Bed 1 (template) — 4 students: Alice, Bob, Cara, Dan" rather than 5 unrelated cards.

- **No scoring against** `activity.answer_key` in M14 yet. The

`answer_key` exists (M11), but the matcher between student-charted events and the key's expected findings is a v7.1 module. Once it lands, the Analysis block gains a rubric-score field per encounter.